Ronald Kouatchoua Djakou

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Education

Bachelor of Science Candidate | **Mathematics** | California Institute of Technology | *Pasadena, CA, USA* | **Graduating: 06/2029**

Experience

Caltech — **Undergraduate Researcher** — **June 2026 - Present**

- Developed a **Python** pipeline that **reduced peak memory usage from 59 GB to 2.5 GB(95.8% reduction) and increased throughput by 400%**. Results were verified to be byte-for-byte identical.
- Designed and implemented an algorithm that **eliminated 80 hours** of manual data labeling while improving the consistency and reproducibility of results
- Developed research algorithms and robust comparison techniques that **sped our research timeline by 2 weeks**.
- All implementations above delivered automations that prior efforts by other researchers failed at.

Caltech — **Undergraduate Researcher** — **April 2026 - June 2026**

- Built a **Rust/WebAssembly** interface for simulation execution and data analysis, enabling researchers to generate, access, and visualize results through a unified interface.
- Independently designed and built **Rust-based** simulation software for cellular automata studies, filled a gap where no existing tooling was available.
- Implemented **Rust-based** data storage and manipulation reducing the number of simulations required over time.

Caltech — **First Year Success Research Institute Fellow** — **July 2025 - August 2025**

- Developed **Python** critical simulations using **QuTiP** and **SciPy** to study quantum-system behavior under small parameter perturbations.
- Used **NumPy** and **SciPy** to process experimental data and **Matplotlib** to visualize simulation and experimental results that were used by researchers.

Projects

Code visualization interface — **June 2026**

- Built a **Rust** parser that converts source repositories into structured JSON representations of project architecture, enabling AI agents to navigate dependencies and code structure.
- Built a **React** frontend in **TypeScript** and **JavaScript** to visualize repository dependencies, definitions, and data structures from the parser's JSON output.

Database for financial data management — **January - February 2023**

- Designed and built a relational database in **C** and **SQL** to manage financial records and track outstanding debtors, **reducing weekly manual reconciliation by 3 hours**.

Skills

Programming languages: Python, TypeScript, C, HTML/CSS, Rust, JavaScript, SQL
Frameworks and libraries: React, WebAssembly.
Scientific computing: NumPy, SciPy, Matplotlib, QuTiP
Human Languages: English, French

Awards & Honors

- Qiskit Global Summer School 2024 - Quantum Excellence
- IBM Quantum Challenge 2024 Achievement